



ATL A3B: Thinking skills

Apply key ideas and facts in new contexts

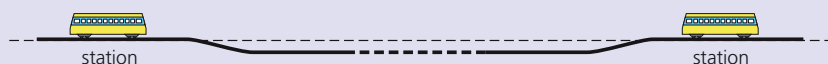
Regenerative braking

Most ways of stopping moving vehicles involve braking systems in which the kinetic energy of the vehicle is transferred to internal energy because of friction in the braking system and with the ground / track. The internal energy is dissipated into the surroundings as thermal energy and cannot be recovered.

The kinetic energy of a long, fast-moving train is considerable. Values of 10^8 J, or more, are not unusual. When the train stops, all of that energy has to be transferred to other forms and, unless the energy can be recovered, the same amount of energy then has to be transferred to accelerate the train again. This is very wasteful, so the train and its operation should be designed to keep the energy wasted to a minimum. One way of doing this is to make sure that large, fast trains stop at as few stations as possible, perhaps only at their origin and final destination.

A lot of research has gone into designing efficient **regenerative braking** systems in recent years, usually involving the generation of an electric current, which can be used to transfer energy to stored chemical changes in batteries, or to operate on-board electrical equipment.

The process of converting the kinetic energy of the train into electrical energy decelerates the train, so that there is much less need for frictional braking. This also reduces the wear on the brakes and the thermal energy transferred to the environment.



■ **Figure A3.38** Possible track profile

◆ Regenerative braking

Decelerating a vehicle by transferring kinetic energy into a form that can be of later use (rather than dissipating the energy into the surroundings). For example, by generating an electric current that charges a battery.

◆ Power, P energy transferred

time taken
or, for mechanical energies,
work done
time taken

◆ **Watt, W** Derived SI unit of power. $1 \text{ W} = 1 \text{ J s}^{-1}$.

Small electric trains, which are often operated underground or on overhead tracks, are a feature of most large cities around the world (Figure A3.37). Such trains usually have stations every few kilometres or less, so regenerative braking systems and other energy-saving policies are very important.



■ **Figure A3.37** The Light Rail Transit trains on the SBS network in Singapore have regenerative braking

When designing a new urban train system, it has been suggested that energy could be saved by having a track shaped as shown in Figure A3.38.

What might be the advantages of such a track profile? Consider the energy transfers taking place as the train moves between stations.

Power

SYLLABUS CONTENT

- Power, P , is the rate of work done, or the rate of energy transfer, as given by: $P = \frac{E}{t} = Fv$.

Power is the rate of transferring energy. When energy is transferred by people, animals or machines to do something useful, we are often concerned about how much time it takes for the change to take place. If the same amount of useful work is done by two people (or machines), the one that does it faster is said to be more *powerful*. (In everyday use the word power is used more vaguely, often related to strength and without any connection to time.)

$$\text{power} = \frac{\text{energy transferred}}{\text{time taken}} \quad P = \frac{E}{t} \quad \text{SI unit: watt, W}$$

$$\text{Alternatively: power} = \frac{\text{work done}}{\text{time taken}} \quad P = \frac{\Delta W}{\Delta t}$$



$1 \text{ W} = 1 \text{ J s}^{-1}$. The units mW, kW, MW and GW are all in common use. (Avoid confusion between work done, W and the unit the watt, W.)

The following are some examples of values of power in everyday life.



■ **Figure A3.39** Transferring energy at a rate of about 250 W

- A 0.0001 W calculator transfers energy at a rate of 0.0001 J every second.
- A 7 W light bulb transfers energy from electricity to light and thermal energy at a rate of 7 J every second.
- A girl walking upstairs may transfer chemical energy to gravitational energy at a rate of about 250 W.
- In many countries, homes use electrical energy at an average rate of about 1 kW.
- A 2 kW water heater transfers energy from electricity to internal energy at a rate of 2000 J every second.
- A typical family car might have a maximum output power of 100 kW.
- A 500 MW oil-fired power station transfers chemical energy to electrical energy at a rate of 500 000 000 J every second.

WORKED EXAMPLE A3.11

Calculate the average power of a 65 kg climber moving up a height of 40 m in 3 minutes.

Answer

$$P = \frac{\Delta W}{\Delta t} = \frac{(mg\Delta h)}{\Delta t} = \frac{(65 \times 9.8 \times 40)}{3 \times 60} = 1.4 \times 10^2 \text{ W}$$

◆ **Resistive force, F** Any force that opposes motion, for example friction, air resistance, drag.

Power needed to maintain a constant speed

It is common for a vehicle to maintain a constant velocity. Under those circumstances, the forward force, F , from the engines is equal in magnitude, but opposite in direction to the magnitude of the total **resistive forces, F** . Then:

Power, P , needed to maintain a constant velocity, v , against the resistive forces, can be determined from:

$$P = \frac{\Delta W}{\Delta t} = \frac{F\Delta s}{\Delta t} = Fv$$



Power needed to maintain a constant velocity, $P = Fv$

WORKED EXAMPLE A3.12

- What average power is needed to accelerate a 1600 kg car from rest to 25 m s^{-1} in 12.0 s?
- What power is needed to maintain the same speed if the resultant resistive force is a constant 2300 N?

Answer

$$\text{a } P = \frac{\Delta W}{\Delta t} = \frac{\text{kinetic energy gained}}{\text{time taken}} = \frac{\frac{1}{2}mv^2}{\Delta t} = \frac{(0.5 \times 1600 \times 25^2)}{12.0} = 4.2 \times 10^4 \text{ W } (= 42 \text{ kW})$$

$$\text{b } P = Fv = 2300 \times 25 = 5.8 \times 10^4 \text{ W } (= 58 \text{ kW})$$

- 33 a** How much useful energy must be transferred to lift twelve 1.7 kg bottles from the ground to a shelf that is 1.2 m higher?
- b** If this task takes 18 s, what was the average useful power involved?
- 34** Estimate the output power of an electric motor that can raise an elevator of mass 800 kg and six passengers 38 floors in 52 s. (Assume there is no counterweight.)
- 35 a** Calculate the average power needed for a cyclist of mass 72.0 kg to accelerate from 8.00 m s^{-1} to 12.0 m s^{-1} in 22.0 s on a horizontal road. Assume that the resistive forces are negligible and the bicycle has a mass of 8.00 kg.
- b** Compare your answer to 6.5 W kg^{-1} (using body mass) for the best athletes.
- 36** A small boat is powered by an outboard motor with a maximum output power of 40 kW. The greatest speed of the boat is 27 knots ($1 \text{ knot} = 1.85 \text{ km h}^{-1}$). Determine the magnitude of the forward force provided by the motor at this speed.
- 37 a** What is the constant speed of a car which has an output power of 22 kW when the resistive forces are 2.0 kN?
- b** What assumption did you make in answering part a?
- 38** What is the output power of a jet aircraft that has a forward thrust of $6.60 \times 10^5 \text{ N}$ when travelling at its top speed of 950 km h^{-1} (264 m s^{-1}) through still air?
- 39 a** Find out which countries of the world have the highest average power consumption per person.
- b** Suggest reasons why they use so much energy.

Efficiency

SYLLABUS CONTENT

- Efficiency, η , in terms of energy transfer or power, as given by: $\frac{E_{\text{output}}}{E_{\text{input}}} = \frac{P_{\text{output}}}{P_{\text{input}}}$.

It is an ever-present theme of physics that, whatever we do, some of the energy transferred is ‘wasted’ (dissipated) because it is transferred to less ‘useful’ forms. In mechanics this is usually because friction, or air resistance, transfers kinetic energy to internal energy and thermal energy. The useful energy we get out of any energy transfer is *always* less than the total energy transferred.

When an electrical water heater is used, nearly all of the energy transferred makes the water hotter and it can therefore be described as ‘useful’, but when a mobile phone charger is used, for example, only some of the energy is transferred to the battery (most of the rest is transferred to thermal energy). Driving a car involves transferring chemical energy from the fuel and the useful energy is considered to be the kinetic energy of the vehicle, although at the end of the journey there is no kinetic energy remaining.

A process that results in a greater useful energy output (for a given energy input) is described as being more efficient. In thermodynamics, **efficiency** is defined as follows:

◆ **Efficiency (thermodynamic), η** Ratio of useful energy (or power) output to the total energy (or power) input.



$$\text{efficiency, } \eta = \frac{\text{useful energy output}}{\text{total energy input}} = \frac{E_{\text{output}}}{E_{\text{input}}}$$

Dividing energy by time to get power, we see that efficiency may also be defined as:



$$\text{efficiency, } \eta = \frac{\text{useful power output}}{\text{total power input}} = \frac{P_{\text{output}}}{P_{\text{input}}}$$

Because it is a ratio of two energies (or powers), efficiency has no units. It is often expressed as a percentage. It should be clear that, because of the principle of conservation of energy, efficiencies will always be less than 1 (or 100%).

It is possible to discuss the efficiency of any energy transfer, such as the efficiency with which our bodies transfer the chemical energy in our food to other forms. However, the concept of efficiency is most commonly used when referring to electrical devices and engines of various kinds, especially those in which the input energy or power is easily calculated. Sometimes we need to make it clear exactly what we are talking about. For example, when discussing the efficiency of a car, do we mean only the engine, or the whole car in motion along a road with all the energy dissipation due to resistive forces?

The efficiencies of machines and engines usually change with the operating conditions. For example, there will be a certain load at which an electric motor operates with maximum efficiency; if it is used to raise a very small or a very large mass it will be less efficient. Similarly, cars are designed to have their greatest efficiency at a certain speed, usually about 100 km h^{-1} . If a car is driven faster (or slower), then its efficiency decreases, which means that more fuel is used for every kilometre travelled.

Car engines, like all other engines that rely on burning fuels to transfer energy, are inefficient because of fundamental physics principles (see Topic B.4). There is nothing that we can do to change that, although better engine design and maintenance can make some improvements to efficiency.

In recent years we have all become very aware of the need to conserve the world's energy resources and limit the effects of burning fossil fuels in power stations and various modes of transport on global warming (see Topic B.2). Improving the efficiency of such 'heat engines' has an important role to play in this worldwide issue.

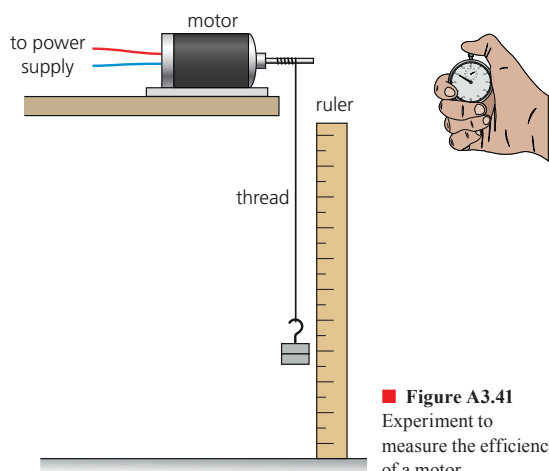


■ **Figure A3.40** Power stations which use natural gas have the greatest overall efficiency ($\approx 60\%$)

WORKED EXAMPLE A3.13

Figure A3.41 shows an experiment designed to measure the efficiency of a small electric motor. Electrical power was supplied to the motor at a rate of 0.72 W. The hanging mass (25 g) went up a distance of 1.12 m in 2.46 s.

- Calculate how much gravitational potential energy was transferred to the mass.
- How much electrical energy was transferred in this time?
- Determine the efficiency of the motor.
- How much energy was dissipated into the surroundings?
- State the forms of this dissipated energy.



Answer

- $\Delta E_p = mgh = 0.025 \times 9.8 \times 1.12 = 0.27 \text{ J}$ (seen on calculator display as 0.2744...)
- power, $P = \frac{\text{energy transferred}}{\text{time taken}}$
energy transferred $= P \times \Delta t = 0.72 \times 2.46 = 1.8 \text{ J}$ (seen on calculator display as 1.7712...)
- efficiency, $\eta = \frac{\text{useful energy output}}{\text{total energy input}} = \frac{0.2744}{1.7712} = 0.15$ (15%)
- $1.7712 - 0.2744 = 1.5 \text{ J}$
- internal energy, thermal energy (+ sound)

We will see in Topic B.5 that the power input to any electric device can be determined from:

$$\text{power} = \text{voltage} \times \text{current}.$$

Inquiry 3: Concluding and evaluating

Concluding

Figure A3.42 represents the apparatus used by a student to investigate the efficiency of a pulley system.

The student measured the forces needed to lift loads of 0.5 kg, 1.0 kg, 4.0 kg and 5.0 kg: 0.6 N, 1.5 N, 7.5 N and 12.7 N. He assumed that the force always moves four times further than the load. His conclusion was that the efficiency of the system was 66%.

- Is this a correct conclusion from the data collected?
- Discuss the quantity and range of readings taken by the student.

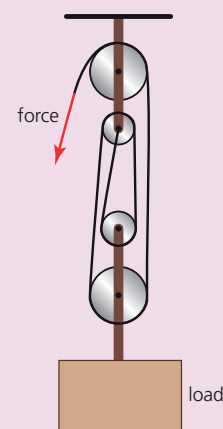


Figure A3.42 Pulley system

Energy density

SYLLABUS CONTENT

◆ **Energy density** The energy transferred from unit volume of fuel (SI unit: J m^{-3}).

◆ **Specific energy** Amount of energy that can be transferred from unit mass of an energy source (SI unit: J kg^{-1}).

► Energy density of the fuel sources.

One of the advantages of fossil fuels is the large amount of energy that can be transferred from each cubic metre of the fuels. Although, nuclear fuels are much more energy dense.

Energy density is the amount of energy that can be transferred from each cubic metre.

SI unit: J m^{-3}

Table A3.4 shows some typical energy densities of energy sources.

When discussing gaseous energy resources, it is more common to use **specific energy**, which is the amount of energy that can be transferred from each kilogram. For example, the specific energy of natural gas is 55 MJ kg^{-1} .

■ **Table A3.4** Some approximate energy densities

Source	Energy density / MJ m^{-3}
Reactor-grade uranium-235	66 000 000 000
coal	43 000
gasoline (petrol)	36 000
crude oil	37 000
ethanol	24 000
wood	15 000
electrical batteries	1000
hydroelectric	1

Tool 2: Technology

Represent data in a graphical form

Use data from the internet to determine specific energies for a variety of different energy sources.

Use a spreadsheet to present the information in the form of a bar chart.

ATL A3C: Research skills

How did you check the reliability of your online sources?

Common mistake

Energy density and specific energy are often confused. Some sources define energy/mass as energy density.

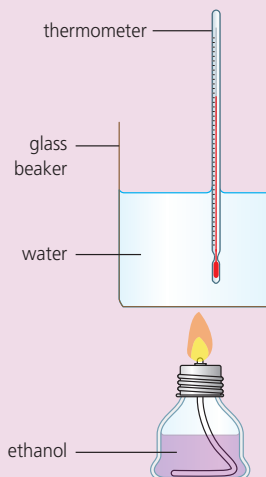
Inquiry 1: Exploring and designing

Designing

Controlling heat losses

Figure A3.43 shows how a student intended to investigate the energy density of ethanol. She knows that for every 4184 J of energy that are transferred to 1.0 kg of water, the temperature will rise by 1.0°C (see Topic B.2).

- 1 Identify the student's choice of dependent, independent and control variables. How would the results be used to calculate energy density?
- 2 Suggest how the experiment (as shown) could be improved. Hint: how much of the energy stored in the fuel is transferred to the water?
- 3 What other energy transfers are taking place?



■ **Figure A3.43** Estimating the energy density or specific energy of a fuel